

Scalability & Future Plan

Date	15 March 2024
Team ID	SWTID-2026-1156
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application runs on a single server instance	Slow response or downtime when many users log in at once	High
2	Database is not optimised for large data (no indexing / caching)	Queries become slow as records grow	Medium
3	Basic authentication only, no multi-factor login	Higher risk of unauthorised account access	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small group of users on a single server	Deploy on cloud with load balancer and auto-scaling to handle thousands of concurrent users
2	Data Storage	Single local database with limited storage	Move to managed cloud database with regular backups, indexing and replication
3	Performance	Every request hits the database directly	Add caching (e.g. Redis), CDN for static files, and optimise API calls and queries
4	Security	Basic username-password login	Add JWT/OAuth, multi-factor authentication, HTTPS everywhere and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment, database optimisation and caching	1 - 2 months	Faster performance and better reliability
Phase 2	Mobile app version and push notifications	3 - 4 months	Wider reach and better user engagement
Phase 3	AI-based recommendations and analytics dashboard	5 - 6 months	Personalised experience and useful insights
Phase 4	Multi-language support and third-party integrations	7 - 9 months	Larger user base and more partnerships